

the real data distribution or causing the model to blur different data modes together.

7. **Disappearance of Real Tail Data** (Shumailov et al., 2023; Wyllie et al., 2024; Shumailov et al., 2024): Sometimes called “coverage collapse” (Zhu et al., 2024), model collapse here occurs when synthetic data leads to the under-representation of data from the tail of the distribution, leading to models that can only handle common cases but fail on rare ones. The disappearance of real tail data can be more subtle and more narrow than the generative model losing all diversity (Def. 4).

8. **Appearance of Hallucinated Data** (Shumailov et al., 2023; Alemohammad et al., 2023; Bohacek & Farid, 2023): Model collapse occurs when the sequence of models begin producing fully-synthetic data not supported by the original real data’s distribution.

We note that the definitions can themselves be loosely clustered into three families (Fig. 1): (i) population risk degrading (Definitions 1, 2 and 3), (ii) distributions deforming from their original shape (Definitions 4, 5, 6, 7, and 8), and (iii) decreasing value from additional data (Definition 5).

2.1. Intra-Paper Definitions Can Cause Confusion

The differences between different definitions of model collapse can be slippery, and understandably, authors sometimes move between them in the course of a paper. However, model collapse is a technical phenomenon that requires definitional rigor to properly characterize. In this section, we use a prominent prior work to demonstrate how easy it can be to slip between definitions. Our intention is not to call out this specific work, but rather demonstrate how a seemingly reasonable treatment of model collapse definitions can have serious implications for interpreting results.

We consider Shumailov et al. (2023), which admirably provides explicit definitions of two different types of model collapse: “We separate two special cases: early model collapse and late model collapse. In early *model collapse* the model begins losing information about the tails of the distribution; in the late *model collapse* the model entangles different modes of the original distributions and converges to a distribution that carries little resemblance to the original one.” These correspond to our Definitions 7 and 6, respectively.

However, the paper presents results on model collapse that fall under different definitions. Firstly, Shumailov et al. (2023) demonstrate Definition 3 in their Sections 4.2 and 4.3. We lightly generalize their results for clarity and generality. We consider repeatedly fitting multivariate Gaussians to data and sampling from the fitted Gaussians. We begin with n real data drawn from a multivariate Gaussian with mean

$\mu^{(0)}$ and covariance $\Sigma^{(0)}$:

$$X_1^{(0)}, \dots, X_n^{(0)} \sim_{i.i.d.} \mathcal{N}(\mu^{(0)}, \Sigma^{(0)}).$$

For model fitting, we compute the unbiased mean and covariance of the most recent data:

$$\hat{\mu}^{(t+1)} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{j=1}^n X_j^{(t)}$$

$$\hat{\Sigma}^{(t+1)} \stackrel{\text{def}}{=} \frac{1}{n-1} \sum_{j=1}^n (X_j^{(t)} - \hat{\mu}^{(t+1)})(X_j^{(t)} - \hat{\mu}^{(t+1)})^T$$

and then draw n new synthetic samples from a Gaussian with the most recently fit parameters. In this model-data feedback loop, Shumailov et al. (2023) proved that as the model-fitting iteration $t \rightarrow \infty$, the population risk as measured by the expected squared Wasserstein distance between the most recent multivariate Gaussian and the original multivariate Gaussian diverges asymptotically:

$$\mathbb{E}[\mathbb{W}_2^2(\mathcal{N}(\hat{\mu}^{(t)}, \hat{\Sigma}^{(t)}), \mathcal{N}(\mu^{(0)}, \Sigma^{(0)}))] \rightarrow \infty.$$

This result demonstrates that the population risk diverges asymptotically, which aligns with neither of the two definitions of model collapse stated at the outset of the paper. Is it possible that the population risk is diverging *because* of one of the two other definitions? Recall that the squared Wasserstein distance between two Gaussians has two terms: a contribution from the means and a contribution from the covariances:

$$\mathbb{E}[\mathbb{W}_2^2(\mathcal{N}(\hat{\mu}^{(t)}, \hat{\Sigma}^{(t)}), \mathcal{N}(\mu^{(0)}, \Sigma^{(0)}))] =$$

$$\underbrace{\|\hat{\mu}^{(t)} - \mu^{(0)}\|_2^2}_{\rightarrow \infty} +$$

$$\underbrace{\text{Tr}(\hat{\Sigma}^{(t)} + \Sigma^{(0)} - 2((\Sigma^{(0)})^{1/2} \hat{\Sigma}^{(t)} (\Sigma^{(0)})^{1/2})^{1/2})}_{\rightarrow 0}.$$

Thus, while the variance collapses and tails do vanish, neither is the cause of the population risk diverging. Rather, the population risk diverges because the sequence of means $(\hat{\mu}^{(t)})_t$ randomly walks away from the ground truth mean $\mu^{(0)}$. This is one example in which a casual reader might fail to notice that while the population risk *is* indeed asymptotically diverging, such undesirable behavior is attributable to neither real data tails disappearing quickly nor to real data modes entangling slowly.

The same paper also demonstrates how definitions of model collapse can go beyond subtle confusion to explicit contradiction. Shumailov et al. (2023) experimentally demonstrate model collapse under a *fourth* definition of model collapse: in their Section 5.2, the authors consider finetuning sequences of OPT-125M language models (Zhang et al., 2022) initially on wikitext2 (Merity et al., 2016) and

then subsequently on the models’ own generated outputs. The authors’ Figure 10 shows that while the population risk (measured by test perplexity) initially increases, it then decreases and converges to a plateau about 12.5% – 50% above the population risk of the first model. Indeed, the authors find that “Over the generations models tend to produce samples that the original model trained with real data is more likely to produce.” We believe that under Definitions 3, 6, and 7, these results suggest that model collapse has not occurred. However, the authors label this result as model collapse because later generations trained on purely synthetic data begin introducing fully synthetic tail data over time: “later generations start producing samples that would never be produced by the original model, i.e., they start misperceiving reality based on errors introduced by their ancestors.” Thus, this appearance of hallucinated data qualifies as model collapse under Definition 8.

When a paper shifts definitions without explicitly acknowledging the change, it creates a cascade of problems undermining scientific clarity. Readers interpret results through the lens of the initially stated definitions, creating a false sense that all discussed phenomena represent the same underlying issue when they may be fundamentally distinct. This leads to misattribution of causes and effects, as demonstrated in the Gaussian example where population risk divergence was incorrectly associated with tail data disappearance rather than mean drift. Such definitional inconsistency fosters overgeneralization of results, hampering cross-study comparisons and potentially prompting inappropriate technical responses or policy decisions. Most critically, when technical concepts like model collapse require precise characterization, unstated definitional shifts prevent the formation of a stable framework for interpreting claims, ultimately contributing to broader confusion about which phenomena deserve concern and how they might be addressed.

2.2. Inter-Paper Definitions Can Cause Confusion

To demonstrate how different researchers can look at the same results and reach different conclusions regarding model collapse, Alemohammad et al. (2023) studied a FFHQ-StyleGAN2 (Karras et al., 2020) trained on synthetic data and found that the population risk as measured by Frechet Inception Distance (FID) (Heusel et al., 2018) increased $2\times$ by the 5th model-fitting iteration and then plateaued. The authors declared this result constituted model collapse because the authors had implicitly defined model collapse as *any* increase in the population risk (Definition 2). Gerstgrasser et al. (2024) then questioned this claim that the models had collapsed, writing, “Figure 7 from Alemohammad et al. (2023) shows that linearly accumulating data (“Synthetic augmentation loop”) causes poor behavior to plateau with the number of model-fitting iterations [...] We believe is that our evidence and their evidence is more

consistent with the conclusion that accumulating data avoids model collapse and does not merely delay it.” This apparent disagreement was because Gerstgrasser et al. (2024) had defined model collapse as asymptotically diverging population risk (Definition 3). Thus, while looking at the exact same figure, the researchers came to differing conclusions because they were operating under different definitions.

2.3. Stating and Adhering to Definitions Improves Clarity and Drives Progress

In this position paper, our intention is not to argue in favor of specific definitions of model collapse or call out authors whose definitions we disagree with. Rather, we hope to emphasize that model collapse is a multifaceted phenomenon: under the same results, model collapse can simultaneously “occur” and “not occur” in different researchers’ opinions; in Appendix A, we include a case study of how confusing and entangled scientific insights can become on account of different definitions and methodologies. This makes building a comprehensive understanding of model collapse difficult, which can be especially concerning to specific communities. For instance, search providers like Google, Bing, and Perplexity may pay an especially high penalty if their models are trained on hallucinated facts, whereas the disappearance of real tail data can disproportionately affect marginalized groups or historically disadvantaged communities (Blodgett et al., 2016; Bender & Friedman, 2018; Noble, 2018; Shah et al., 2020; Jo & Gebru, 2020; Koenecke et al., 2020; Bender et al., 2021; Hutchinson et al., 2021; Ji et al., 2023). By clarifying different definitions of model collapse and adhering to those definitions, researchers can build a better understanding of model collapse with greater nuance for what causes different failures modes and what actions can be taken to prevent each. Together, these definitions can form a “model collapse profile” which can characterize the different ways in which models worsen over time. Research on the harms of synthetic data should explicitly note the relevant aspects of the model collapse profile they study.

3. Realistic Conditions for Studying Model Collapse

Our goal is to understand likely outcomes as humanity pre-trains future frontier AI systems on web-scale datasets containing a mixture of real and synthetic data. We focus on pre-training both because the literature has (Shumailov et al. (2023)’s “What will happen to GPT-n once LLMs contribute much of the language found online?”) and because pre-training’s tremendous capital and operating expenses render missteps extremely costly. Motivated by this goal, we posit trends that we believe describe the current trajectory of pre-training practices of frontier AI systems:

1. **Increasing Pre-training Compute:** The total floating point operations used for pre-training has been rapidly increasing. For example, Meta pre-trained Llama 1 using 2k GPUs, Llama 2 using 4k GPUs and Llama 3 using 16k GPUs (Goyal, 2024), and OpenAI recently announced a \$500B initiative to increase compute capacity of the United States (OpenAI & SoftBank, 2025), a fraction of which will be allocated for pre-training.
2. **Increasing Pre-training Data:** The amount of data used has been rapidly increasing. For example, Meta’s series of Llama language models were pre-trained on increasing amounts of data: 1.4 trillion tokens for Llama 1, 2 trillion tokens for Llama 2, and most recently, 15 trillion tokens for Llama 3. While pre-training data will inevitably max out, recent estimates place the total number of available language pre-training tokens at 1 quadrillion (1000 trillion) tokens (Villalobos et al., 2024).
3. **Increasing Quality of Pre-training Data:** Over time, pre-training data is becoming increasingly higher quality on account of pre-training data teams developing better filtering techniques to ensure high-quality training data (Gao et al., 2020; Penedo et al., 2024; Li et al., 2024). Moreover, whatever synthetic data is shared online is increasingly higher quality since models are improving over time (Kiela et al., 2021; 2023; Maslej et al., 2024).
4. **Synthetic Data Accumulating Alongside Real Data:** Real data are not deleted en masse after each iteration of model pre-training. When synthetic data are generated and released online, they amasses alongside prior data and new real data.
5. **Decreasing Proportion of Real Data:** The fraction of real data relative to total (real plus synthetic) data is decreasing over time. However, whether the fraction of real data will asymptote to zero is unclear, a point we will return to later.

We believe that researchers and policy makers interested in potential societal implications of model collapse should focus on research that adhere to these conditions as faithfully as possible (with the obvious caveat that computational budgets limit research).

4. Key Dimensions of Consideration for Model-Data Feedback Loops

4.1. Propagation of Data Over Time

Early work that sounded the alarm about model collapse (Martínez et al., 2023; Alemohammad et al., 2023; Bohacek & Farid, 2023; Shumailov et al., 2023; Briesch et al., 2023;

Bertrand et al., 2023) assumed that data propagate in a particular way: after training a model, all existing data are deleted, new data are sampled from the new model, and the next model is trained solely on this freshest synthetic data. Subsequent authors called this the *replace* paradigm (Gerstgrasser et al., 2024; Kazdan et al., 2024b; Dey & Donoho, 2024) because data are entirely replaced after each model-fitting iteration. When data are replaced, researchers demonstrated multiple harmful outcomes: variances collapse, real data tails disappear, population risk diverges and so on (Fig. 3 left). This particular assumption of how data propagate over time is highly unrealistic: **After a model finishes training, the entire internet is not deleted, nor is the next model necessarily trained solely on its predecessor’s outputs.** Rather, a more realistic assumption is that synthetic data from each model *accumulates* on the internet alongside real data and past synthetic data such that all can be used for training the next model. To some, these differences might seem insignificant, but each produces vastly different asymptotic behavior in terms of population risk: Gerstgrasser et al. (2024) showed empirically and Kazdan et al. (2024b) and Dey & Donoho (2024) showed mathematically that population risk diverge if data are replaced, but population risk does not diverge if data instead accumulate (Fig. 3 right).

A slightly different but perhaps more realistic data propagation assumption is that real and synthetic data accumulate, but future models are trained on a downsampled proportion of the total available data (Kazdan et al., 2024b). This *accumulate-subsample* paradigm represents a middle ground between replace and accumulate: while test loss appears to stabilize, no analytic theory has been proven in this case to date. Individual beliefs about which data paradigm is most reflective of reality can influence perceptions of how model collapse will unfold in the future. However, to our knowledge, non-population risk-based notions of model collapse have not been connected to assumptions about how data propagate, leaving an important question open.

4.2. Proportion of Real Data Over Time

ChatGPT alone produces 1/1000 of all words produced by humanity each day (Altman, 2024), and as these models proliferate, over time, future generative models could produce vastly more data than humanity for training future models. Thus, the proportion of real data on the future internet plays a crucial role in the debate over the effects of model collapse. Bertrand et al. (2023) claimed that the population risk will not asymptotically diverge so long as the proportion of real data remains lower-bounded above 0 (in addition to other conditions). Relatedly, in Dohmatob et al. (2024b)’s view, *any* synthetic data causes “a critical degradation” to future models, writing model collapse “generally persists even when mixing real and synthetic data, as long as the